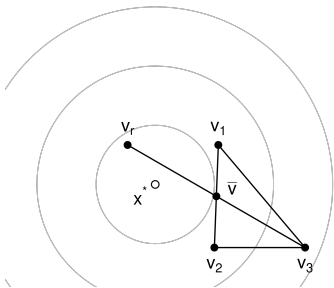


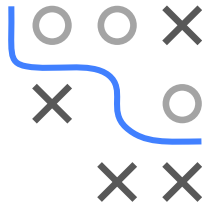
Optimization in Machine Learning

Nelder-Mead method



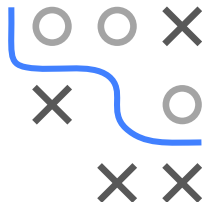
Learning goals

- General idea
- Reflection, expansion, contraction, shrinking
- Advantages & disadvantages
- Examples



NELDER-MEAD METHOD

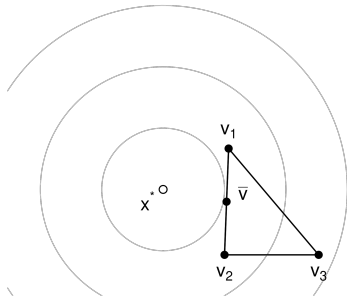
- Derivative-free method $\Rightarrow$ heuristic ► Nelder and Mead 1965
- Based on d -simplex, defined by $d + 1$ points:
 - $d = 1$ interval
 - $d = 2$ triangle
 - $d = 3$ tetrahedron
 - ...
- Idea: replace worst vertex by a better point on the line through it and the centroid of the remaining vertices; if none is found, shrink simplex towards best vertex
- Unrelated to the simplex method of linear programming



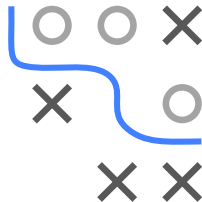
NELDER-MEAD METHOD

2. Compute centroid without worst point

$$\bar{\mathbf{v}} = \frac{1}{d} \sum_{i=1}^d \mathbf{v}_i.$$



Simplex with computed centroid $\bar{\mathbf{v}}$



NELDER-MEAD METHOD

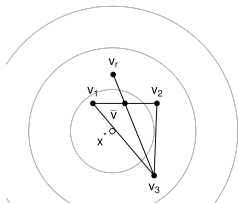
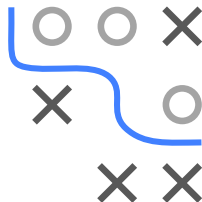
- Case 3: $f(\mathbf{v}_r) \geq f(\mathbf{v}_d) \Rightarrow$ Contraction with $0 < \gamma < 1$

Note: default value for contraction coefficient: $\gamma = 1/2$

- Case 3a: $f(\mathbf{v}_d) \leq f(\mathbf{v}_r) < f(\mathbf{v}_{d+1}) \Rightarrow$ Outside contraction:

$$\mathbf{v}_{oc} = \bar{\mathbf{v}} + \gamma(\mathbf{v}_r - \bar{\mathbf{v}})$$

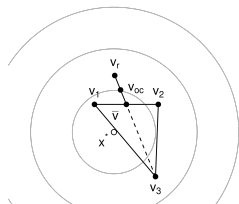
If $f(\mathbf{v}_{oc}) \leq f(\mathbf{v}_r)$: accept $\mathbf{v}_{oc}$ and discard $\mathbf{v}_{d+1}$



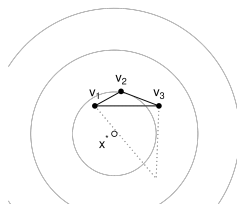
Reflection (2nd iteration)

with $\rho = 0.4$:

$$f(\mathbf{v}_2) \leq f(\mathbf{v}_r) < f(\mathbf{v}_3)$$



Case 3a: outside
contraction with $\gamma = 1/2$



Case 3a: acceptance of
 $\mathbf{v}_{oc}$

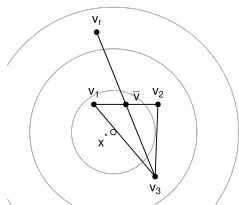
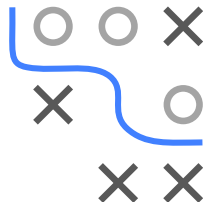
NELDER-MEAD METHOD

- Case 3 (continued):

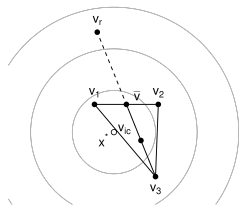
- Case 3b: $f(\mathbf{v}_r) \geq f(\mathbf{v}_{d+1}) \Rightarrow$ Inside contraction:

$$\mathbf{v}_{ic} = \bar{\mathbf{v}} + \gamma(\mathbf{v}_{d+1} - \bar{\mathbf{v}})$$

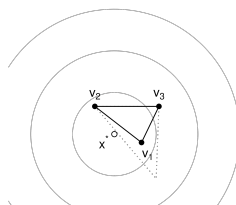
If $f(\mathbf{v}_{ic}) < f(\mathbf{v}_{d+1})$: accept $\mathbf{v}_{ic}$ and discard $\mathbf{v}_{d+1}$



Reflection (2nd iteration)
with $\rho = 1$: $f(\mathbf{v}_r) \geq f(\mathbf{v}_3)$



Case 3b: inside
contraction with $\gamma = 1/2$



Case 3b: acceptance of
 $\mathbf{v}_{ic}$

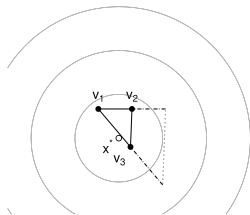
NELDER-MEAD METHOD

- Case 3 (continued):
 - If contraction point is not accepted, i.e., $f(\mathbf{v}_{oc}) > f(\mathbf{v}_r)$ in case 3a or $f(\mathbf{v}_{ic}) \geq f(\mathbf{v}_{d+1})$ in case 3b: shrink entire simplex towards $\mathbf{v}_1$ (Shrinking)

$$\mathbf{v}_i = \mathbf{v}_1 + \sigma(\mathbf{v}_i - \mathbf{v}_1), \quad i = 2, \dots, d + 1$$

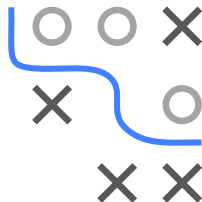
with $0 < \sigma < 1$.

Note: default value for shrinking coefficient: $\sigma = 1/2$



Hypothetical visualization of case 3: shrinkage with $\sigma = 1/2$ (if $f(\mathbf{v}_{ic}) \geq f(\mathbf{v}_3)$)

4. Repeat all steps until stopping criterion met



NELDER-MEAD SUMMARY

Advantages:

- No gradients needed
- Robust, often works well for non-differentiable functions

Drawbacks:

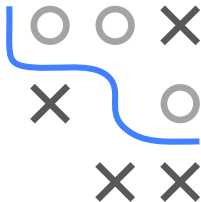
- Relatively slow (not applicable in high dimensions)
- Best point need not improve in each step, only the mean of vertex values $\frac{1}{d+1} \sum_{i=1}^{d+1} f(\mathbf{v}_i)$ is guaranteed to decrease (except in shrink steps) ► Nocedal and Wright 2006
- No guarantee for convergence to local optimum / stationary point

Visualization:

<http://www.benfrederickson.com/numerical-optimization/>

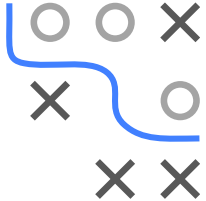
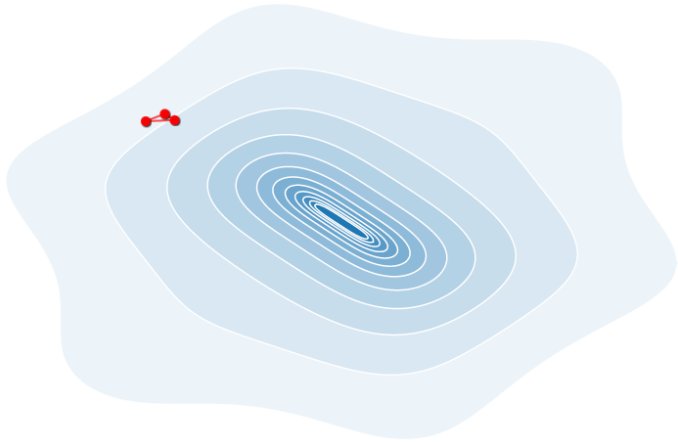
Note: Nelder-Mead is default method of R function `optim()`

If gradient is available and cheap, L-BFGS is preferred



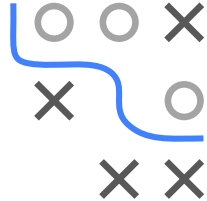
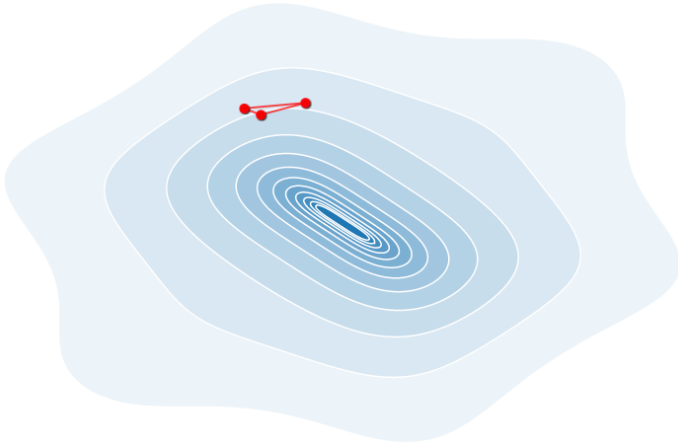
NELDER-MEAD VISUALIZATION IN 2D

$$\min_{\mathbf{x}} f(x_1, x_2) = x_1^2 + x_2^2 + x_1 \cdot \sin x_2 + x_2 \cdot \sin x_1$$



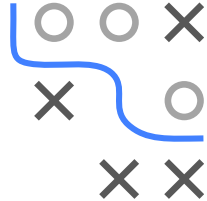
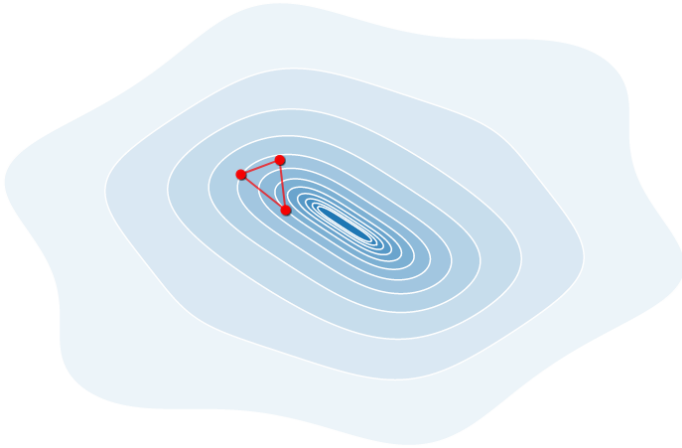
NELDER-MEAD VISUALIZATION IN 2D

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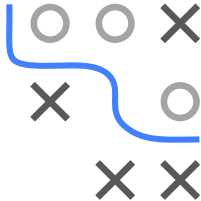
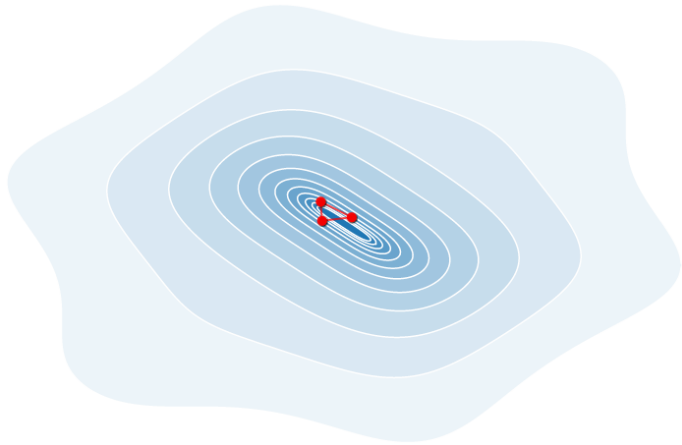
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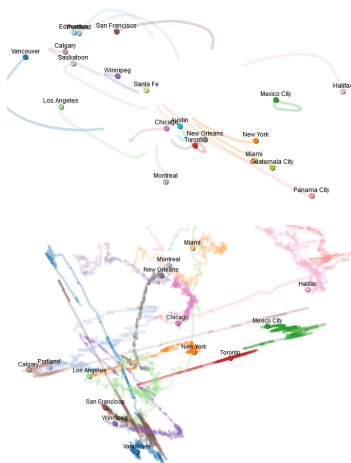
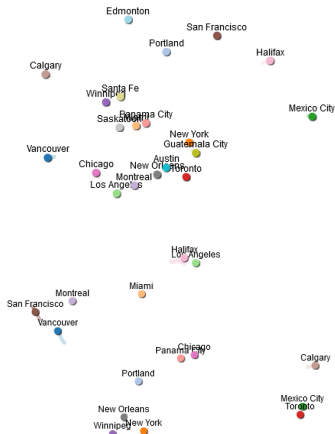
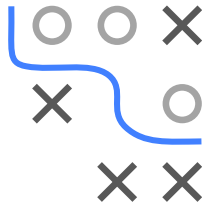


NELDER-MEAD VISUALIZATION IN 2D

$$\min_{\mathbf{x}} f(x_1, x_2) = x_1^2 + x_2^2 + x_1 \cdot \sin x_2 + x_2 \cdot \sin x_1$$



NELDER-MEAD VS. GD



Nelder-Mead in multiple dimensions: Organize points (US cities) to keep predefined mutual distances. For 10 cities, gradient descent (top) converges well for a suitable learning rate. Nelder-Mead (bottom) fails to converge, even after many iterations.

